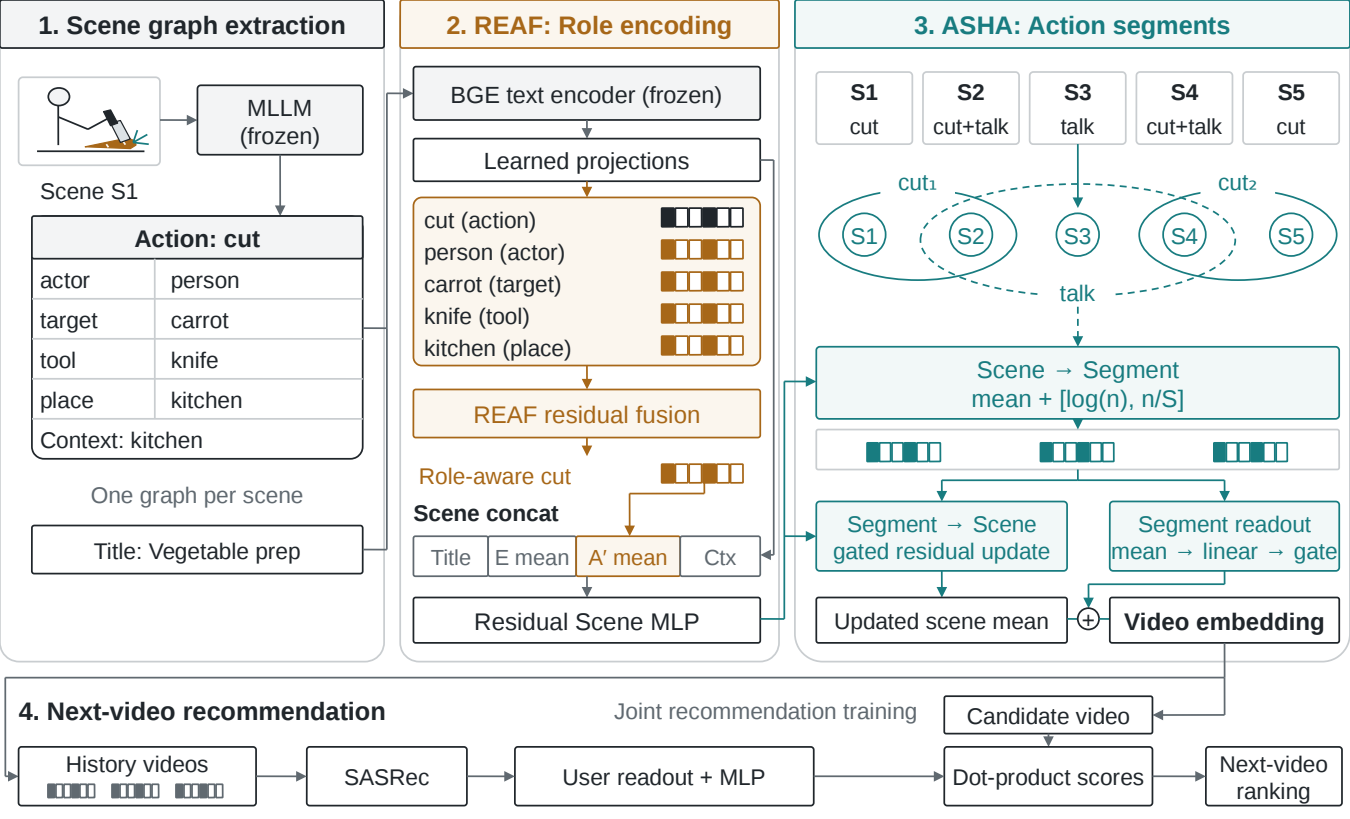

1. Scene graph extraction
MLLM (frozen)
Scene S1
Action: cut
actor
person
target
carrot
tool
knife
place
kitchen
Context: kitchen
One graph per scene
Title: Vegetable prep
2. REAF: Role encoding
BGE text encoder (frozen)
Learned projections
cut (action)
person (actor)
carrot (target)
knife (tool)
kitchen (place)
REAF residual fusion
Role-aware cut
Scene concat
Title
E mean
A′ mean
Ctx
Residual Scene MLP
3. ASHA: Action segments
S1 cut
S2 cut+talk
S3 talk
S4 cut+talk
S5 cut
cut1
cut2
S1
S2
S3
S4
S5
talk
Scene → Segment mean + [log(n), n/S]
Segment → Scene gated residual update
Segment readout mean → linear → gate
Updated scene mean
Video embedding
4. Next-video recommendation
Joint recommendation training
History videos
SASRec
User readout + MLP
Candidate video
Dot-product scores
Next-video ranking